

Document outline:

- Anything written in black is priority
- Anything written in blue is for future priority (unless easy to add) – so don't worry about it for now but let's discuss them once the main app is largely build, unless again you think it is easy to add at the current moment.
- Validation criteria are in red

Section 1: Main thoughts and ideas

So, the main idea for this app is a tool for bird banders (or other folks doing different types of research on birds) to manage + collect their data easily in the field and back in office. It will be ideal for any project doing banding, and we should look at it more as a “band deployment manager” rather than a “session data manager,” if that makes sense. Since data is centered around USGS BBL-issued bands, and less centered around session data, we should think of band numbers anchoring all additional data.

Overall, I think it would be easiest to think of the data collection/management in 4 different categories:

Category 1: Session Data. This will be the metadata for different sessions but does not necessarily need to be tied to the banding data specifically. Session Data will be collected at the start of each session and will ideally have a referenceable session ID that can be linked on the banding data. Ideally, we could create different prompts or templates depending on what type of protocol we are doing but let's just start with MAPS/non-maps which will be identical protocols for now. The overall structure and workflow for this interface will be outlined in Section 2.

Category 2: Project Location Data. Since we (and most banders) will be banding in multiple known project locations (I will be banding at a minimum at Galindo Creek, possibly Mitchell Canyon, at Lindsay Wildlife (records will be centered around where they are released, however), at NASA Mountain View, and a few other areas, it may be easiest to create a location manager to store location data. This location data will be linked to Session Data and Banding Data (for each deployment + encounter) via a Bander Location ID. Ideally, as well, for rehabbed birds – we may need to re-think this as I band them at Lindsay Wildlife but often update the record to be where they were released, but I will need to get back to you – so for now, I will link it to the Lindsay Wildlife location and include release location in the notes until we can determine a better way down the road. The overall structure and workflow for this interface will be outlined in Section 2.

Category 3: Band Inventory. This will be the band inventory manager. As a requirement of the USGS BBL, we must keep track of every band we get issued (they do as well). Each band's fate must be accounted for, and each band only has a single fate. The band manager would be used to keep track of existing bands and hold data as to whether it's been deployed or not. Ideally, I would like to be able to click on a band number and be brought to the band profile + encounter history, but that is something that is not needed immediately. As well, this would ideally also track auxiliary markers and their deployments alongside the aluminum bands. The band inventory will be tightly linked to the banding data and will need to automatically update when a band is deployed. The overall structure and workflow for this interface will be outlined in

Section 2.

Category 4: Banding Data Collection. This will be the main page we are working off in the field, which you have solidly gotten a good first draft on. This will be an interface to either create a new record for a newly deployed band, or create a new record for a new encounter (aka recapture) with either a local band (our band that we deployed) or a foreign band (from another banding station (band number will not be within our band inventory)). The interface can (and should) be the same for all situations, but will provide the ability to enter data, have the data validate based off of a set of known ranges in measurements and criteria, and ideally self-validate based off written scenarios of contradicting data entered in multiple categories (difficult to explain, so not initially needed). Ideally, as well, I would like the ability to link banding data to photos or to any addendum datasheets. Once all mandatory fields are collected, this interface will submit the record to a master database of which all records will be referenceable and exportable, and will (immediately after submitting) update the band inventory + records. The overall structure and workflow for this interface will be outlined in Section 2.

Section 2: Overall flow and vision for appearance and structure

When entering BirdNerd, to start, it would be nice to have selection buttons for the 4 categories: Collect Project Location Data, Collect Session Data, Band Inventory, and Banding Data Collection. Additionally, we might need to create a 5th selection to view data and/or export data, but not sure if this would be imbedded in another area or its own area (you decide, you're the expert). Additionally, it might be good to also have a button to select initially for me to provide comments or report bugs. Additionally down the line, it would be sick to also have a photo log for wing photos linked to the banding data, to have the ability to fill out and link the Empidonax flycatcher forms and the *Selasphorus* hummingbird forms, and other types of data collected (like color band resighting) – but again that's down the line!

a) Project Location Data interface. This will be relatively simple, but we will need the ability to add new locations, as well as view and manage existing locations (including export). When we log locations onto the BBL portal, we usually collect the following information:

- BBL Location ID; ##### (this number is issued by the BBL after they accept your location data. We won't need to input this # immediately, but it would be nice to be able to add it after BBL issues it to us)
- Bander Location ID: this is a 4-letter ALPHA code for our locations, selected by us. This code will be what we use to link location data to both the Session data and the banding data. Currently, we have Galindo Creek Banding stations as "GCBS" and Lindsay Wildlife as "LMEX."
- Coordinates (decimal (like 37.94008, -121.97276)) which could ideally be taken using the devices built-in GPS if possible, but could be pre-entered ahead of time.
- Country (dropdown list; OR self-populating following coordinates entry)
- State/Province (dropdown list that changes depending on the Country selection; OR self-populating following coordinates entry)
- Bander Remarks (optional notes section)
- Net/Trap Inventory. This will be the ability to log info for nets, traps, nests, etc. at each Project Location. I see this as a referenceable table rather than something that needs to be linked to the Banding Data or Session Data (although Banding Data and Session Data

will have entries detailing our which nets/traps/nests were opened/accessed, and which net/trap/nest new encounters or deployments came from) unless linking is easier. For each trap/net/nest, I would like to log what type it is (12m mist-net, 10m mist-net, walk-in trap, etc.), where it is located specifically within the Project Location (coordinates), or other information depending on what type of trap is selected. I'd say if and when we get to this level of detail, I will type out the full list of what is needed.

b) Collect Session Data interface. This interface will have a log of old sessions, as well as the ability to create a new session. The log of old sessions will need to be referenceable, modifiable, and exportable. Additionally, I would like sessions to be linked to the Banding Data. If helpful to view, our current Session Datasheet can be found [here](#). Ideally, I would like to start by selecting a protocol (Mist-Netting, Trapping, Nest-box Monitoring, etc) which would pull up different forms to fill out with different criteria.. but for now, I am just writing the scenario for mist-netting (MAPS/Non-MAPS) data collection. When creating a new mist-netting session, we will need to collect the following data points:

- Session ID: some type of referenceable ID that we can use to link to the Banding Data.
- Protocol (dropdown: MAPS, Non-MAPS, burrowing owl banding, Rehabbed-Bird Banding, Saw-whet Owl Banding, etc) (ideally, in the future, this selection will happen before the form pops up and again will have different forms for different protocols, but for now we need a way to trace the sessions to a certain protocol)
- Date (mm/dd/yyyy) preferentially an auto-select using device data
- MAPS Period (see [this doc](#) for periods as well help understanding effort data)
- Project Location (dropdown selection of the 4-letter ALPHA codes logged into Project Location data)
- Effort Data including all opened net number(s) and their associated open & close times. I am not sure the best way to do this part but essentially, we need to select which nets were opened and include which times they were opened or closed, with any remarks. On an ideal day, we open all 10 nets at the same time and close them at the same time, but this rarely happens. So, we need to be prepared to collect the data in a way that reflect that, and design this collection section that allows that – I'm just not sure how. We use these data to track how many “net hours” we complete, which is a measure of effort used when running analyses later on. We also need to include remarks when nets are opened or closed in a non-standard fashion. Ideally, in the remarks area, we will have a dropdown selection to make this easier (Closed early due to wind, closed early due to predators being present, opened late due to low temps, etc).
- Net hours completed (this will be auto-populating after completing the effort information)
- Weather data. This will need two collections – one at start of session “Open” and one at end of session “Close”. This will include temperature (Celsius), Wind (Beaufort), Cloud Cover (%), and Precipitation (select: fog, thick fog, drizzle, rain).
- Personnel list (optional dropdown list but I can see it being a pain if we add new people often, would start with Master Bander/Sub-permittee (Hallie Daly, Julie Woodruff, Tatyana Soto-Bartzi, Joanna van Dyk, Lucas Corneliussen)
- Notes + Observations.
- Optional results section, which would be an auto-populating summary after completing

the session, which would auto-populate from the banding data that is linked to the session data.

c) Band Inventory interface. This interface will need to accomplish a few things, including adding new bands, modification of band information, and viewing of band status. When viewing this interface, it would be nice to be able to see a quick inventory overview where it details by band size + type how many have been deployed versus how many are remaining. We will also need, in this section, to have the full list of all issued bands. For each band, we have the Band Number (XXXX-XXXXX(X)**), Band Size, Type, Quantity Deployed, and Quantity Remaining. As well, it would be nice to be able to click on each band number and see the entire history including the initial deployment and any subsequent recaptures, [resighting in the case we add a collection method for band-resighting efforts, or public encounters from folks who re-capture or find bands and report it straight to the BBL \(this will likely have to be entered manually\)](#). I need this inventory to auto-update following data submission in the Banding Data, but also be modifiable and exportable. [Additionally, please note that on rare occasions bands are sometimes changed, or additional bands are added. In these scenarios, an old band is removed and replaced with or joined by a new band. Both band numbers will be associated with the full records/history if that makes sense.](#)

**In general, bands are issued by strings of 100 in a continuous series identified by a prefix (####) and a suffix (#####(#)) of 5-6 numbers

[Down the line \(May?\), when we start doing color banding, we will also have an inventory of “Auxiliary Markers.” As of right now, we are looking at colored bands with 1-2 letters and 1-2 numbers. Each auxiliary marker is deployed *with* a BBL band. Because they will always be linked to a BBL band, it may be easier to not put auxiliary markers in the inventory and only have a method of adding those data to the Banding Data records during collection, but we will cross that bridge when we get there!](#)

d) Banding Data Collection interface. This is where we spend 90% of our time! There is a LOT of data that needs to be taken, and a LOT of validation that will need to occur here, so put on your boots and let's get into it! ((note anything with * will be REQUIRED to be filled before submission)

[Overall, I think some things: one it may be easiest to first select if it's a New, Recapture, or Unbanded collection as it may change some validations and may change exactly what we need to collect... food for thought as we work through this.](#)

[Additionally, the order that we collect the data/layout may change after reviewing different stages of this app– so keep that in mind.](#)

-*Band Number: XXXX-XXXXX (dropdown of list of all bands; if a recaptured band with no records, have “FOREIGN RECAPTURE” pop up in large words; if band with history, I would like a small box or option to view old records that include date of each encounter, activity of each encounter (“Deployment” or “Recapture” or [“Resighting”](#)), Sex, Age, WRP, and Disposition – see example table below:

Date	Activity	Sex	Age	WRP	Disposition
01/01/2025	Deployment	U	6 - SY	FCF	
04/09/2026	Recapture	M	7 - ASY	DPA	S

In addition, please add the ability to select UNBANDED instead of a band number. Lastly, once a band number is selected, please have the band size and band type display on the screen for reference + verification.

-*Code (selection: 1 – New, U – Unbanded, R – Recapture, F – Foreign Recapture, 4 – Destroyed, 5 – Band Replaced, 6 – Added to, 8 – Band Lost, X – Banding Mortality) **Validation:** note it is impossible to report a recapture as New, Destroyed, Band Lost; however you can report a recapture as any other code. If someone selects a band number that has been used, maybe only the options that are possible appear, OR if contradictory selections are made and error pops up – whatever is easiest. Additionally, new bands should only be paired with band numbers that are in inventory as unused or available.

-*Species (a full table will be provided for you, however have the selection be the common name (i.e. California Towhee) and please have the ALPHA code auto populate as appropriate (or you can tie it together like you have currently) **Validation** certain species get a specific band size, however sometimes we gauge their legs and they are outside of the standard size. In these cases, I would like an error or something to pop up. I would like the data entry person to be able to select a box or override (or questions pop up “Did you gauge the leg?”) the error to acknowledge the discrepancy. In the case they acknowledge, I ideally would like a comment to auto populate in the notes section that states “Leg gauged.” I can provide you the table of species and their standard band sizes.

(I suggest organizing the following collection items into a “Age and Sex” category)

-*Age (select: U – Unknown, L – Local, HY – Hatch Year, AHY – After Hatch Year, SY – Second Year, ASY – After Second Year, TY – Third Year, ATY – After Third Year)

-*How Aged (selection: BP – Brood Patch, CC – Characteristics/Measurements, CL – Cloaca, EG – Egg in Oviduct, EY – Eye Color, FB – Fault Bar, FF – Flight Feather Condition/Wear, IC – Inconclusive/Conflicting, LP – Molt Limit, MB – Mouth/Bill, MR – Active FF Molt, NA – Not Attempted, NF – Local Bird, NL – No Molt Limit, NN – Nestling in Nest, PL – Body Plumage, RC – Recapture Records, SK – Skull, **OT – Other (*require a note be added before submission)** (how aged not required if Age is U)

-WRP (selection: FPJ, FCJ, FPX, FCX, FPF, MFPP, FCF, MFCE, HFCE, AFCE, FPA, MFPA, FCA, MFCA, FPS, MFPS, FCS, MFCS, FCU, FPU, SPB, MSPB, SCB, MSCB, SPA, MSPA, SCA, MSCA, SPS, MSPS, SCS, MSCS, SCU, SPU, TPB, MTPB, TCB, MTCB, TPA, TCA, TCU, TPU, DPB, DCB, DPA, DPS, DCS, DCU, DPU, UPB, UCB, UCA, UPA, UPU, UUU, **Other (*require a note be added before submission)**).

-*Sex (select: M, F, U)

-*How Sexed (select: BP – Brood Patch, CC - Characteristics/Measurements, CL – Cloaca, EG – egg in Oviduct, EY – Eye Color, MB – Mouth/Bill, NA – Not Attempted, PL – Body Plumage,

TL – Tail Length, WL = Wing Length, OT – Other (*require a note be added before submission)
(how sexed not required if Sex is U)

(I suggest organizing the following collection items into a “Condition” category)

-Skull (selection: 0 - None, 1 - Trace, 2 - < 1/3, 3 - Half, 4 - > 2/3, 5 – Almost Complete, 6 – Fully Complete, 8 – Invisible) (make required if SK is selected in How Aged) (ideally would like to provide an info icon which can be selected and a reference to the skulling code visual-aid could be provided)

-Brood Patch (select: 0=None, 1=Smooth, 2=Vascular, 3=Heavy, 4=Wrinkled, 5=Molting)
(validate and give error if both Sex is selected as M and either 3 or 4 are selected here)

-Cloacal Protuberance (select: 0=None, 1=Small, 2=Medium, 3=Large) (validate and give error if both Sex is selected as F and 1-3 are selected here)

-Fat (select: 0=None, 1=Trace, 2=Light, 3=Half, 4=Filled, 5=Bulging, 6= Gr. Bulging, 7=V. Excess)

-Body Molt (0=None, 1=Trace, 2=Light, 3=Medium, 4=Heavy)

-Flight Feather Molt (select: N=None, A=Adventitious, S=Symmetric, J=Juv. Growth)

-Flight Feather Wear (select: 0=None, 1=Slight, 2=Light, 3=Moderate, 4=Heavy, 5=Excessive)

-Juvenile Body Plumage (select: 3=Full, 2=>1/2, 1=<1/2, 0=None)

(I suggest organizing the following collection items into a “Molt Limits and Plumage” category, and making it a table with drop downs)

PCovs	SCovs	PP	SS	Tert	Rec	Body Plum	Non-Feather

drop downs selection: J=Juvenile, L=Juv.&Form., F=Formative, B=Basic, R=Juv.&Basic, M=Mixed Basic, A=Alternate, N=Non-Juvenile, U=Unknown

(down the line, I would like be able for the app to self-validate, however its going to take a lot of work and coding of situations... so lets leave it be for now)

(I suggest organizing the following collection items into a “Morphometrics and Status” category)

-Wing (mm) (## entry) (validate against known range for species, table will be provided)

-Tail (mm) (## entry) (validate against known range for species, table will be provided)

-Tarsus (mm) (##.## entry) (validate against known range for species, table will be provided)

-Exposed culmen (##.## mm) (entry) (validate against known range for species, table will be provided)

-Other (mm) (##.## entry) (require note)

-Mass (g) (##.# entry) (validate against known range for species, table will be provided)

-*Status (many different ones, but common for now; selection: 300 – healthy + banded, 700 – Rehabbed and Banded, 500 – Sick/Injured/Stressed and banded (require disposition to be filled and note), 301 – healthy, banded, + color banded, 318 – healthy, banded, and blood sample, 319 – healthy, banded, auxiliary, blood sample, 333 – healthy, banded, taken from artificial nest structure, 334 – healthy, banded, auxiliary, taken from artificial nest structure, 380 – healthy, satellite/cell/gps transmitter, “---” – Mortality (require note), Other (require note) (do NOT require Status entry if unbanded)

-Disposition (select: M – Malformed, O – Old/Healed Injury, I – Illness/Disease, S – Stress/Shock, E – Eye Injury, T – Tongue Injury, W – Wing Injury, B – Body Injury, L – Leg Injury, P – Predation, D – Dead (will eventually want cross-validations so that Status and Disposition match up)

(I suggest organizing the following collection items into a “Additional Information” category)

-*Session ID (once selected, have it display date and Location)

-*Bander ID

-*Capture Time (hh:mm)

-*Release Time (hh:mm) (tap to auto fill time on current device)

-*Net/Trap (manual entry until we can figure out a drop-down from what is entered in session data)

-Note (have it auto-populate if certain selection above are selected, and allow the ability to add extra in addition; notes aren’t required unless certain validations are triggered above)

-Feather Pull? (select box) (backend: no selection = NO, selection = YES)

-Blood Sample? (select box) (backend: no selection = NO, selection = YES) (if selected, validate Status is 518)

e) Master Data/Export. Ideally, the export files will be different based off what agency we need to submit it to (the excel export will look different for our records versus for submission to BBL versus submission to IBP, CDFW, etc.) so having the ability to export straight into the excel layout we need will save a lot of time. This will take awhile to figure out, though, as some agencies have different codes for different entry items.

e) Report bugs/give feedback. Just a way we can communicate + give feedback if others try it out.

f) Photo Log. This ideally would attach photos to the banding record, be a visual database, and be able to export photos where the files name themselves.

g) Datasheet Addendums. We have a variety of datasheets with supplemental info that we occasionally collect for certain records. I can create a folder of these, but somehow we’d need to

fill them out during the data collection process for the band, append them to the record, and be able to export them with a naming convention.

Question for Ken:

-for the backend data collection, would it be easiest for me to provide you with an excel? [Here](#) is the master sheet we are currently using. Note the different tabs for the different agencies submittals, but I am still setting this up. It may be best if we walk through the back end together sometime? Let me know